UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2007 question paper

5054 PHYSICS

5054/04 Paper 4 (Alternative to Practical), maximum raw mark 30

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1 (a) (i) 10 to 20 oscillations [1]

(ii) T too small / time measured larger / gives time on stopwatch about 10s / not too long to take readings / large number may lose count / error in T is 1/N error in t / good comment on reaction time

NOT just makes T more accurate [1]

(b) check for error in timing/ practice increases competence / **average** gives more accurate time / increases sf in T [1]

(c) paper clip moving fastest / time when passing fiducial marker

NOT makes T more accurate [1]

(d) oscillations too fast to count/ time too small to measure [1]

(e) axes, correct way round, labelled quantity and unit
scales; more than ½ page, sensible
5 points plotted accurately $\pm$ ½ small square
best fit curve drawn, neatly [4]

[Total: 9]

2 (a) (i) normal drawn perpendicular to mirror where ray arrives [1]

(ii) 59^{o} to 60^{o} unit required [1]

(b) (i) reflected ray drawn accurately from mirror and through P_3 and P_4 [1]

(ii) reflected ray drawn accurately from mirror and through P_5 and P_6 [1]

(iii) 40 ± 1 [1]

(iv) 2 ecf (b) (iii) / 20 **no unit** [1]

(v) repeat experiment for different value of z
additional detail, e.g. compares new c to original c
at least two additional values of z
plots graph of y against z [2]

[Total: 8]

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3 **(a)** quantities: temperature and time **NOT** temperature change

units: °C and seconds (s) or minutes (min not m) [2]

(b) **(i)** y-axis labelled temperature or temperature change, x-axis labelled time
(allow symbols for quantities)

(ii) correct curve shape for y-axis label

(iii) for y = temperature, values 90° and 20° marked on temperature axis,
and line starts at 90 °, ends at 20 °
OR for y = temperature change, value 70 ° marked on temperature change axis,
and line from 0 to 70 °
unit required on axes labels or on values on axes
ignore curve shape [3]

(c) temperature continuously changing / only one temperature at each time [1]

(d) any two clear practical details e.g.
at least 1/3 thermometer immersed
avoid parallax when reading thermometer (any explanation must be correct)
use of two people
heat above 90° and start stopwatch as temp reaches 90°
read from top of mercury meniscus
mercury column in line with scale
stir water
large number of readings taken
stopwatch close to thermometer
external factors constant [2]

[Total: 8]

4 **(a)** **(i)** newton meter / spring balance / force meter

(ii) 4.6 to 4.9 1 dp only

(iii) 1.5 or 1.6 [3]

(b) 6.9 cm ecf (a) (ii) and (iii) **NOT** one sf [1]

(c) water on the block will change the weight / time needed to dry cube [1]

[Total: 5]

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